3 Fig. 3.1 shows two children balanced on a seesaw. A seesaw is a length of wood which rotates about a central pivot.

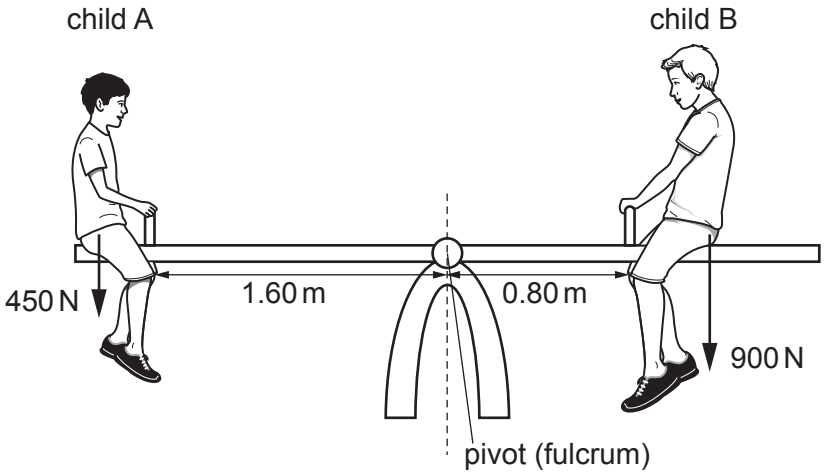


Fig. 3.1

(a) Child B moves 0.050 m further away from the pivot.

(i) Explain why the seesaw rotates clockwise.

.....

..... [1]

(ii) Child A puts on a backpack and the seesaw now balances.

Calculate the mass of the backpack.

mass = ... kg [3]

(b) The concrete floor under the seesaw is replaced with a rubber floor. A child falls from the seesaw and experiences an impulse when they hit the floor.

(i) Define impulse.

.....

..... [1]

(ii) Explain how the rubber floor reduces injury to the child.
Use ideas about impulse, force, momentum and time in your answer.

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.....

.....

..... [3]

[Total: 8]